src/dynamics.py · combine_payload MMA3001 DeepONet project

combine_payload

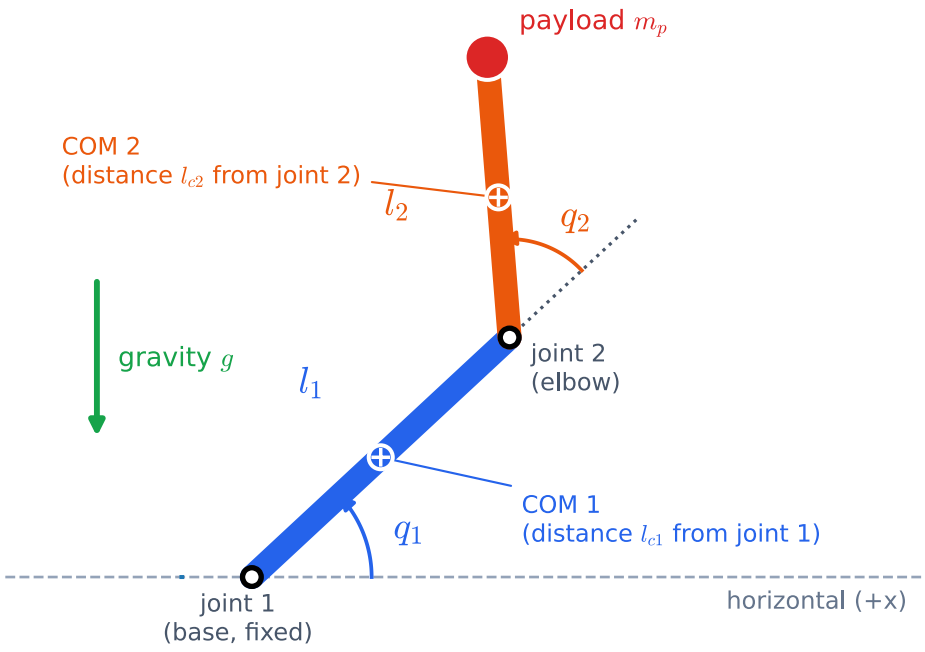
Folding a tip payload into link 2: mass, centre of mass and inertia, derived from scratch

In this document

1. The question
2. Toolbox: COM, inertia, parallel-axis theorem
3. Derivation of the three outputs
4. Why replacing the real body is exact
5. Worked example
6. Code-maths dictionary
7. Sanity checks

The set-up: a two-link robot arm in a vertical plane

Everything in this project describes one physical object: a planar robot arm with two rigid links, two motorised joints, bolted to a wall at the origin. It moves in a *vertical* plane, so gravity matters. At the tip of link 2 we may clip on an extra weight (the **payload**). Each joint has a little friction that slows it down in proportion to how fast it spins.



The arm. Blue = link 1, orange = link 2. The angle q_1 is measured from the horizontal; the angle q_2 is measured from the *straight-on extension* of link 1 (dotted), not from the horizontal. The dots with a white cross are the links' centres of mass (COM).

Conventions (fixed everywhere in the project): the state is $x = [q_1, q_2, \dot{q}_1, \dot{q}_2]$; gravity points in the $-y$ direction; potential energy is zero at the height of joint 1; positive angles and torques are anticlockwise.

Symbol	Meaning	Value in the code	Unit
$l_1,\ l_2$	link lengths (joint to joint)	0.5, 0.4	m

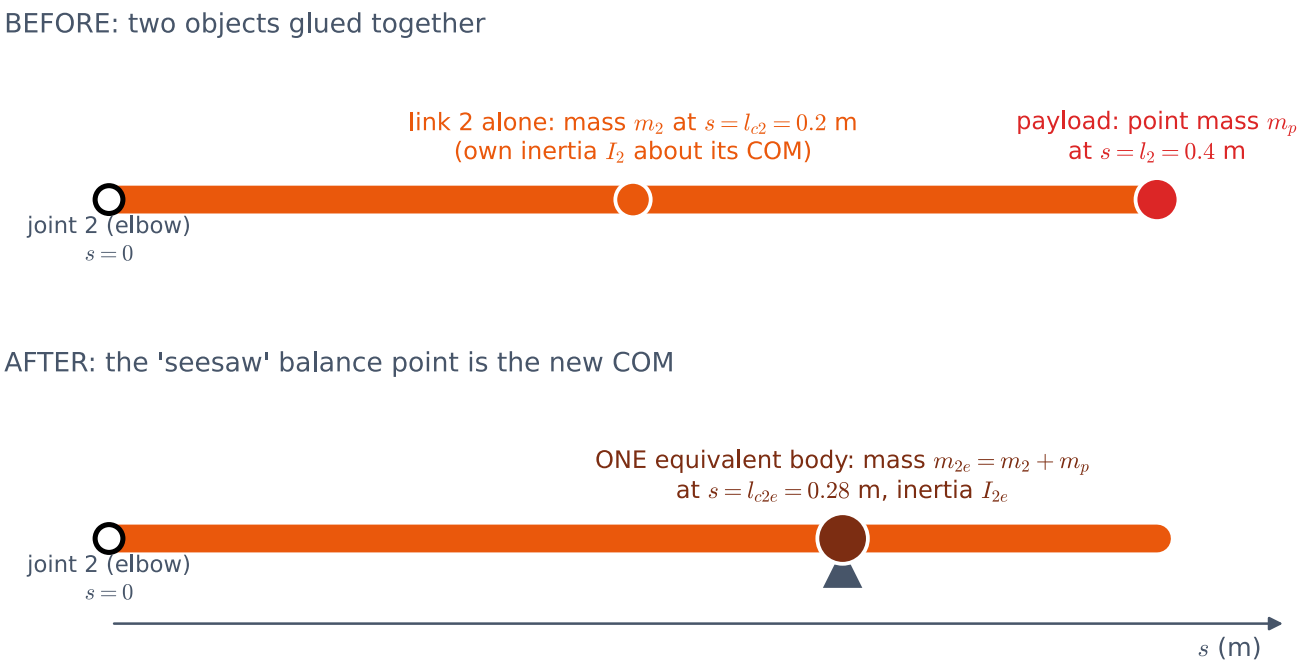
$m_1,\ m_2$	link masses (without payload)	2.0, 1.5	kg
$l_{c1},\ l_{c2}$	distance from a link's joint to its own centre of mass (COM)	0.25, 0.2 (middle of each link)	m
$I_1,\ I_2$	moment of inertia of a link about its own COM	$ml^2/12$ (uniform rod)	kg m^2
g	gravitational acceleration	9.81	m/s^2
m_p	point-mass payload at the tip of link 2 (varies per sample)	0 to 1 (dataset), up to 1.5 out-of-range	kg
b	viscous joint friction coefficient (varies per sample)	0.05 to 0.5	N m s/rad
$q_1,\ q_2$	joint angles: q_1 from the horizontal, q_2 relative to link 1	state	rad
$\dot{q}_1,\ \dot{q}_2$	joint angular velocities (a dot means d/dt)	state	rad/s
$\tau_1,\ \tau_2$	motor torques applied at joints 1 and 2 (the inputs)	$A\sin(2\pi ft+\phi)$	N m

1 The question this function answers

All the equations of motion we will meet (the mass matrix, the Coriolis term, gravity) are written for an arm whose links are each described by three numbers: a **mass**, a **centre-of-mass position** and an **inertia**. But our arm may carry an extra weight m_p clipped onto the tip of link 2. That payload is *not* part of the link, and it is a different kind of object (a tiny blob, not a rod). Do we have to re-derive everything for "a rod plus a blob"?

No. If two rigid things are firmly attached to each other, they behave as *one* rigid body. So we only need to find the three numbers $(m,\ l_c,\ I)$ of that single combined body and then reuse every formula unchanged. That is all `combine_payload` does:

```
def combine_payload(m2, lc2, I2, l2, m_p):
    m2_eff  = m2 + m_p
    lc2_eff = (m2*lc2 + m_p*l2) / (m2 + m_p)
    I2_eff  = I2 + m2*(lc2_eff - lc2)**2 + m_p*(l2 - lc2_eff)**2
    return m2_eff, lc2_eff, I2_eff
```



Replacing link 2 + payload by one equivalent body. Top: the real situation, with link 2 (orange) and a heavy blob at the tip (red). Bottom: the single body that the rest of the code sees. Its mass is the sum; its centre of mass sits where the whole thing would balance on a knife-edge (the grey triangle); its inertia is derived below. Numbers drawn for $m_p = 1$ kg.

Three questions remain, and we answer them one by one: where is the new centre of mass? what is its new inertia? and why is it legitimate to replace the real thing by this one body?

2 Toolbox: the three ideas you need

2.1 Centre of mass = the balance point

The centre of mass (COM) of an object is the point where you could balance it on a knife-edge. Gravity pulls on every little piece, but the object behaves as if all the weight were at that one point. For a few point masses m_i at positions s_i along a line (here s is the distance measured from joint 2 along link 2), the balance point s^* is where the "twists" cancel, like on a seesaw:

$$\sum_i m_i \left(s_i - s^*\right) = 0 \quad \Longrightarrow \quad s^* = \frac{\sum_i m_i s_i}{\sum_i m_i}.$$

That is simply a **mass-weighted average** of the positions. (Check: two equal masses balance in the middle; a heavier mass pulls the balance point towards itself.)

2.2 Moment of inertia = resistance to being spun

Mass measures how hard it is to *push* something; the **moment of inertia** I measures how hard it is to *spin* something. Take a small mass m on a weightless stick of length r, spinning at angular speed ω (rad/s). Its speed is $v = \omega r$, so its kinetic energy is

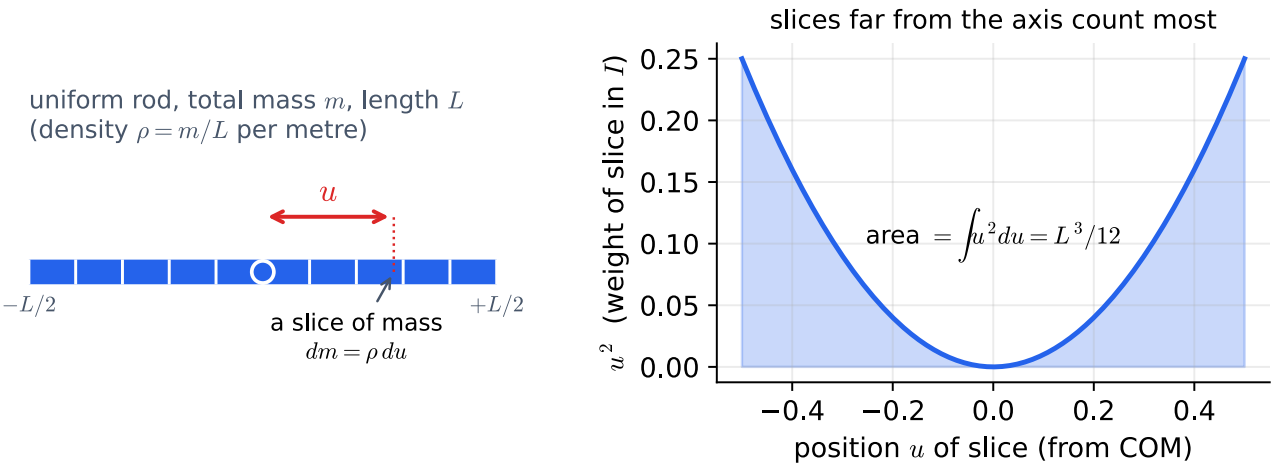
$$\tfrac{1}{2} m v^2 = \tfrac{1}{2}\left(m r^2\right)\omega^2 \equiv \tfrac{1}{2} I \omega^2, \qquad I = m r^2.$$

Note the square: mass far from the axis is *much* harder to spin than mass close to it. For a collection of masses we just add: $I = \sum_i m_i r_i^2$, and for a continuous body the sum becomes an integral, $I = \int r^2 \, dm$.

Example: a uniform rod about its middle. Cut the rod (mass m, length L) into slices of width du at position $u \in [-L/2,\, L/2]$ from the centre. Each slice has mass $dm = \rho\, du$ with $\rho = m/L$.

$$I_{\text{cm}} = \int_{-L/2}^{L/2} u^2\, \rho\, du = \rho \left[\frac{u^3}{3}\right]_{-L/2}^{L/2} = \frac{m}{L} \cdot \frac{2}{3} \cdot \frac{L^3}{8} = \boxed{\frac{mL^2}{12}}.$$

This is where the defaults in the code come from: `I1 = 2.0*0.5**2/12` and `I2 = 1.5*0.4**2/12` treat each link as a uniform rod. (Careful: the subscript "cm" means the axis goes through the *centre of mass*.)



Inertia of a rod about its centre. Left: the rod cut into slices. Right: each slice contributes $u^2\, dm$; the area under u^2 gives $L^3/12$, hence $I_{\text{cm}} = \rho L^3/12 = mL^2/12$.

2.3 The parallel-axis theorem (moving the axis)

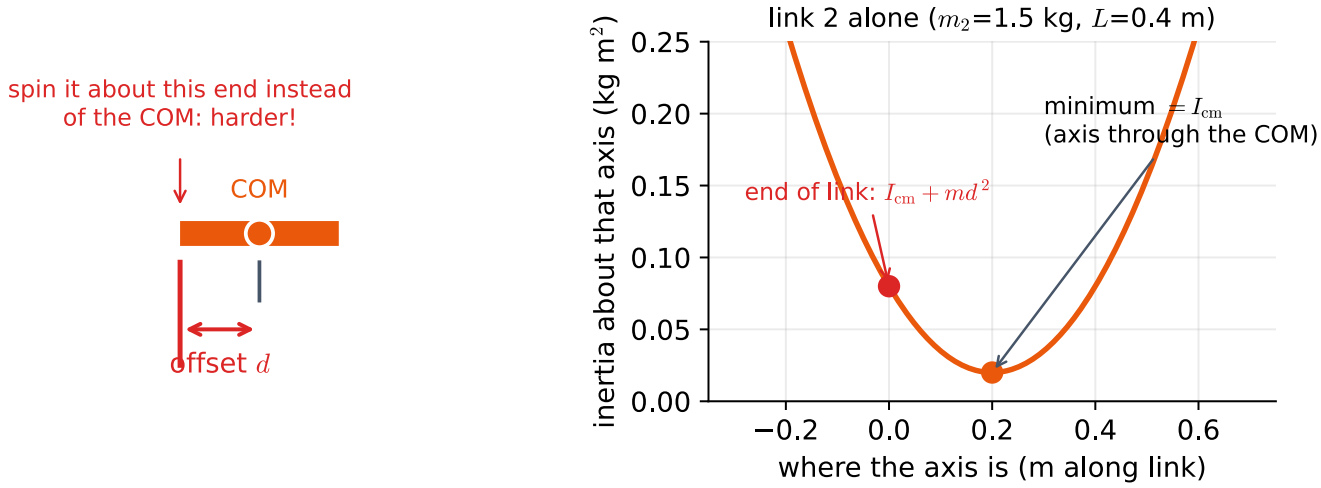
We need the inertia about the *new* COM, but link 2's inertia I_2 is about *its own* COM, a different point. The **parallel-axis theorem** converts between them. If I_{cm} is the inertia of a body (mass M) about an axis through its COM, then about a parallel axis a distance d away,

$$\boxed{I = I_{\text{cm}} + Md^2}$$

Proof (1D version, all we need). Let each piece of the body sit at position u measured from the COM. By the definition of COM, $\int u\, dm = 0$ (positive and negative offsets cancel). A new axis at distance d from the COM sees each piece at distance $u + d$:

$$I = \int (u+d)^2 dm = \underbrace{\int u^2 dm}_{I_{\text{cm}}} + 2d \underbrace{\int u\, dm}_{=0} + d^2 \underbrace{\int dm}_{M} = I_{\text{cm}} + Md^2. \qquad \blacksquare$$

So the inertia is smallest about the COM and grows like a parabola as the axis moves away (graph below). A point mass has no size, so its own $I_{\text{cm}} = 0$ and its whole inertia is the Md^2 part.



Parallel-axis theorem. Left: the same rod, but the axis is moved to its end, a distance $d = L/2$ from the COM. Right: inertia versus axis position for link 2 on its own: a parabola whose lowest point is I_{cm} at the COM.

3 Derivation of the three outputs

Measure position s along link 2 from joint 2 (the elbow). The two ingredients are:

	mass	position of its COM	inertia about its own COM
link 2	m_2	$s = l_{c2}$	I_2
payload (point mass at the tip)	m_p	$s = l_2$	0 (no size)

Step 1: the mass

Masses simply add: $m_{2e} = m_2 + m_p$.

Step 2: the centre of mass

Apply the mass-weighted average of section 2.1 to the two objects:

$$l_{c2e} = \frac{m_2\, l_{c2} + m_p\, l_2}{m_2 + m_p}.$$

This is the line `lc2_eff = (m2*lc2 + m_p*l2) / (m2 + m_p)`. Look at the two limits: for $m_p \to 0$ it gives l_{c2} (nothing changed); for $m_p \to \infty$ it gives l_2 (the payload dominates and the balance point moves to the tip).

Step 3: the inertia about the new COM

We want the inertia of the *whole* thing about the point $s = l_{c2e}$. Inertias add (the sum $\sum mr^2$ is just a sum over all pieces), so treat the two objects separately and use the parallel-axis theorem on each:

- **Link 2.** Its own COM is at l_{c2}, but our axis is at l_{c2e}, so the offset is $d_1 = l_{c2e} - l_{c2}$. Contribution: $I_2 + m_2 d_1^2$.
- **Payload.** It sits at l_2, offset $d_2 = l_2 - l_{c2e}$. A point mass has $I_{\mathrm{cm}} = 0$, so the contribution is $0 + m_p d_2^2$.

$$\boxed{I_{2e} = I_2 + m_2\,(l_{c2e} - l_{c2})^2 + m_p\,(l_2 - l_{c2e})^2}$$

This is exactly the third line of the function (the comment in the code, "parallel axis theorem", refers to this step). Note that the squares make both corrections *positive*: adding a payload and shifting the axis can only

increase the inertia.

4 Why can we replace the real thing by one body? (the subtle part)

The rest of the model uses the combined triple $(m_{2e}, l_{c2e}, I_{2e})$ in the same formulas that were derived for a simple link. This is correct because those formulas only ever depend on three properties of a body: its **mass** (in the gravity term), the **position of its COM** (heights and speeds of the COM) and its **inertia about the COM**. The reason is **König's theorem**: the kinetic energy of any rigid body that translates and rotates is

$$T = \tfrac{1}{2} M v_{\mathrm{cm}}^2 + \tfrac{1}{2} I_{\mathrm{cm}}\, \omega^2 .$$

Proof sketch. Write the velocity of each piece as $v_i = v_{\mathrm{cm}} + \dot{u}_i$, where u_i is its position relative to the COM. Then $\tfrac{1}{2}\sum m_i |v_i|^2 = \tfrac{1}{2} M v_{\mathrm{cm}}^2 + v_{\mathrm{cm}} \cdot \sum m_i \dot{u}_i + \tfrac{1}{2}\sum m_i |\dot{u}_i|^2$. The middle term is $v_{\mathrm{cm}} \cdot \frac{d}{dt}\sum m_i u_i = 0$ because $\sum m_i u_i = 0$ by the definition of COM. In a rotation $|\dot{u}_i| = \omega |u_i|$, so the last term is $\tfrac{1}{2}\omega^2 \sum m_i u_i^2 = \tfrac{1}{2} I_{\mathrm{cm}} \omega^2$. ■

Likewise gravitational potential energy is $\sum m_i g y_i = M g\, y_{\mathrm{cm}}$ (again only mass and COM height). Since T and V are all that Lagrange's method uses, any object with the same $(M, \mathrm{COM}, I_{\mathrm{cm}})$ gives exactly the same equations of motion. The substitution is *exact*, not an approximation.

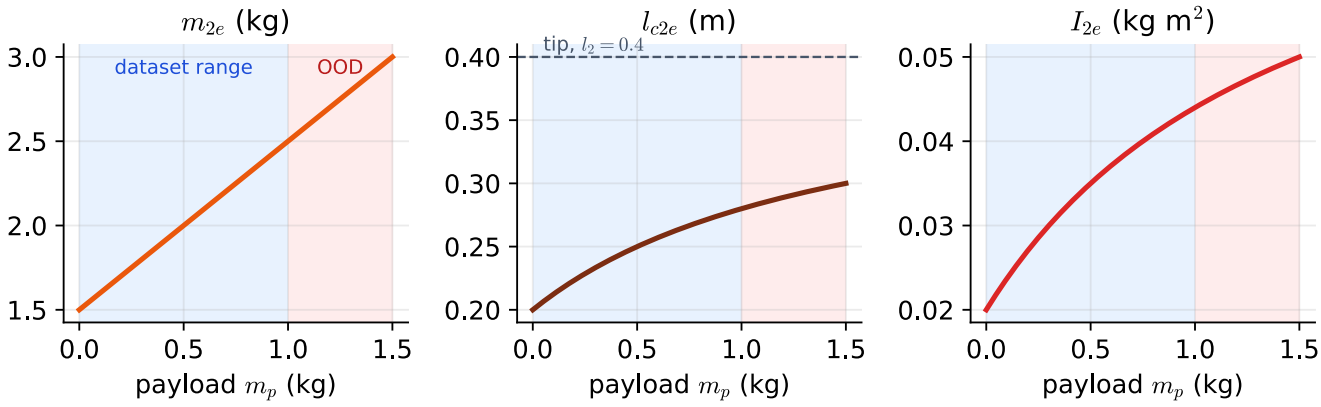
> **NOTICE**
> A useful product appears: $m_{2e}\, l_{c2e} = m_2 l_{c2} + m_p l_2$ (just multiply Step 2 by $m_2 + m_p$). This is the "total weight-times-distance" of link 2 plus payload, and it is what shows up in the gravity term. It means the payload's effect on gravity is simply linear in m_p.

5 Worked example with real numbers

Default arm: $m_2 = 1.5$ kg, $l_{c2} = 0.2$ m, $l_2 = 0.4$ m, $I_2 = m_2 l_2^2/12 = 0.02000$ kg m². Add a payload $m_p = 1.0$ kg.

quantity	calculation	result
m_{2e}	$1.5 + 1.0$	2.5000 kg
l_{c2e}	$(1.5 \cdot 0.2 + 1.0 \cdot 0.4)/2.5000 = 0.700/2.5000$	0.2800 m
offsets	$d_1 = l_{c2e} - l_{c2}$, $d_2 = l_2 - l_{c2e}$	0.0800 m, 0.1200 m
I_{2e}	$0.02000 + 1.5\,(0.0800)^2 + 1.0\,(0.1200)^2 = 0.02000 + 0.00960 + 0.01440$	0.04400 kg m²

The payload nearly doubles the inertia of link 2 (0.02 to 0.044) even though it adds only 67% more mass, because it sits at the far end (the r^2 effect again). These numbers are exactly what `combine_payload(1.5, 0.2, 0.02, 0.4, 1.0)` returns (I checked).



How the equivalent body changes with the payload (default arm). Left: mass grows linearly. Middle: the COM slides from 0.20 m towards the tip at 0.40 m but never reaches it. Right: inertia grows steeply. Blue band = payload range used for training data (0 to 1 kg); red = the out-of-distribution test range (1 to 1.5 kg).

6 Code ↔ maths dictionary

code	maths	meaning
`m2_eff = m2 + m_p`	$m_{2e} = m_2 + m_p$	total mass
`lc2_eff = (m2*lc2 + m_p*l2)/(m2+m_p)`	$l_{c2e} = \frac{m_2 l_{c2} + m_p l_2}{m_2 + m_p}$	balance point (weighted average)
`m2*(lc2_eff - lc2)**2`	$m_2 d_1^2$	parallel-axis term of link 2
`m_p*(l2 - lc2_eff)**2`	$m_p d_2^2$	parallel-axis term of the point mass
`I2 + ... + ...`	I_{2e}	inertia of the glued body about the new COM

7 Sanity checks (the project's unit tests do exactly these)

- $m_p = 0$ must return the original (m_2, l_{c2}, I_2): then $d_1 = 0$ and the payload terms vanish. ✓
- $m_p \to \infty$ must put the COM at the tip, $l_{c2e} \to l_2$, and $m_{2e} \to m_p$ (see `test_huge_payload_moves_com_to_tip`). ✓
- Symmetry check: the COM must lie between l_{c2} and l_2 (a weighted average always lies between its inputs), so $d_1 \geq 0$ and $d_2 \geq 0$ and the inertia can only rise.

CHECK YOURSELF

1. A 0.5 kg payload is put on the default arm. Where is the new COM? *Answer:* $(1.5 \cdot 0.2 + 0.5 \cdot 0.4)/2.0 = 0.25$ m.
2. Why is the payload term $m_p(l_2 - l_{c2e})^2$ and not $m_p l_2^2$? *Answer:* we want the inertia about the *new COM*, not about the elbow; the distance of the payload from that axis is $l_2 - l_{c2e}$. (The elbow-axis version is used later inside the mass matrix, as $m_{2e} l_{c2e}^2 + I_{2e}$.)
3. Does the payload change m_1, l_{c1} or I_1? *Answer:* No: it hangs on link 2, so only link 2's triple changes. Link 1 feels the payload only through the motion it must provide, which the mass matrix handles.